

A Zero-Knowledge Adaptive AI Agent with Secure Multi-User Memory and Controlled Knowledge Evolution using RAG

Abstract

Recent advances in Large Language Models (LLMs) have enabled intelligent conversational agents capable of generating high-quality responses across multiple domains. However, most existing AI assistants operate as stateless systems that neither retain user-specific knowledge nor adapt safely over long-term interactions. Moreover, naïve memory integration introduces risks such as cross-user data leakage, hallucinated personalization, and uncontrolled knowledge accumulation.

This project proposes the design and implementation of a Zero-Knowledge, Adaptive, Self-Learning AI Agent that begins with no prior assumptions about users and incrementally learns personalized knowledge through controlled interactions. The system employs multi-user memory isolation to ensure strict separation of user data, and a safe knowledge evolution framework that validates, scores, updates, and selectively forgets learned information over time. To ground responses in factual and user-specific context without retraining the underlying model, the agent leverages Retrieval-Augmented Generation (RAG) using vector-based memory retrieval.

The proposed architecture integrates structured user profiles, long-term vector memory, and short-term session context while enforcing safety constraints such as confidence tracking, contradiction detection, and memory unlearning. The system is evaluated using quantitative and qualitative metrics measuring personalization accuracy, memory relevance, safety, and isolation effectiveness. This work demonstrates a scalable and domain-agnostic approach to building adaptive AI agents suitable for academic, healthcare, enterprise, and other personalized deployment environments.

Proposed Development Timeline (16–18 Weeks)

1. Literature survey, problem analysis, requirement finalization
2. Memory model design (zero-knowledge, confidence, isolation)
3. Core agent logic & knowledge evolution rules
4. RAG pipeline implementation (embedding + retrieval)
5. Multi-user isolation & authentication
6. Safety mechanisms (contradiction detection, unlearning)
7. Evaluation, metrics computation, testing
8. Documentation, report writing, demo preparation

Evaluation Metrics

1. **Personalization Accuracy:** Measures how well the agent adapts to individual users.
2. **Memory Relevance Score:** Evaluates quality of retrieved memories in RAG.
3. **Memory Isolation Integrity:** Ensures no cross-user leakage.
4. **Knowledge Evolution Stability:** Tracks how often memories are corrected or invalidated.
5. **Contradiction Resolution Rate:** Measures system's ability to detect and resolve conflicting facts.
6. **Zero-Knowledge Compliance:** Checks that no personalization occurs without prior learning.

References

1. <https://arxiv.org/abs/2005.11401> (Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, Lewis, P., Perez, E., et al. (2020))
2. <https://aclanthology.org/2021.acl-long.479/> (Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering, Izacard, G., & Grave, E. (2021))
3. <https://arxiv.org/abs/1410.5401> (Neural Turing Machines, Graves, A., Wayne, G., & Danihelka, I. (2014))
4. <https://ieeexplore.ieee.org/document/8673865> (Continual Lifelong Learning with Neural Networks: A Review, Parisi, G. I., et al. (2019))
5. <https://ieeexplore.ieee.org/document/9519395> (Machine Unlearning, Bourtole, L., et al. (2021))
6. <https://aclanthology.org/P18-1205/> (Personalizing Dialogue Agents: I Have a Name I Like to Be Called, Zhang, S., et al. (2018))
7. <https://aclanthology.org/2021.emnlp-main.357/> (Retrieval Augmentation Reduces Hallucination in Conversation, Shuster, K., et al. (2021))
8. <https://www.nowpublishers.com/article/Details/TCS-042> (The Algorithmic Foundations of Differential Privacy, Dwork, C., & Roth, A. (2014))
9. <https://dl.acm.org/doi/10.1145/3442188.3445922> (On the Dangers of Stochastic Parrots, Bender, E. M., et al. (2021))
10. <https://arxiv.org/abs/1702.01932> (A Knowledge-Grounded Neural Conversation Model, Ghazvininejad, M., et al. (2018))